Department of Engineering Science
School of Science & Technology



CES 440 Data Communications
Ch.2 The Physical Layer Part I

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OUTLINE

PART I

- **Functions of Physical Layer**
- **Theoretical Basis for Physical Layer**
- **Guided Transmission Media**
- **Terrestrial Wireless Transmission**
- **Communication Satellite**

PART II

- **Public Switched Telephone System**
- **Mobile Telephone System**
- **Cable Television**

FUNCTIONS OF PHYSICAL LAYER

- **Physical layer is responsible for**
  - Transmitting information over the physical medium
    - Make sure that when a '1' is sent out at Tx, a '1' is received at Rx.
  - Activation and deactivation of physical connections.
- **Physical layer specifies interface characteristics**
  - Mechanical interface.
    - E.g. number of pins of the connector, the shape of the connector, etc.
  - Electrical interface.
    - E.g. volts used to represent '1', how long should a bit last, etc.
  - Procedural interface.
    - E.g. the sequence of events to activate/deactivate a physical medium.
- **Example physical layer standards**
  - RS-232, V.92, X.21

FUNCTIONS OF PHYSICAL LAYER

- **The capacity (maximum transmission rate) of the Tx medium is limited by many factors**
  - Limited bandwidth
    - Spectrum is always a precious resource in communication system.
  - Thermal noise
    - Caused by the random movement of molecules in the electrical equipments.
  - Interference
    - E.g., other electrical devices, motors, other users in the system, etc.
  - Multipath fading (Only in wireless communication system)
    - The distortion of the signal due to reflection, refraction and diffraction of electro-magnetic waveform in wireless environment.
- **How to cope with/fight against this constraints and impairments to improve the system capacity as well as communication quality.**
  - Channel coding, modulation, multiplexing/demultiplexing, etc.

OUTLINE

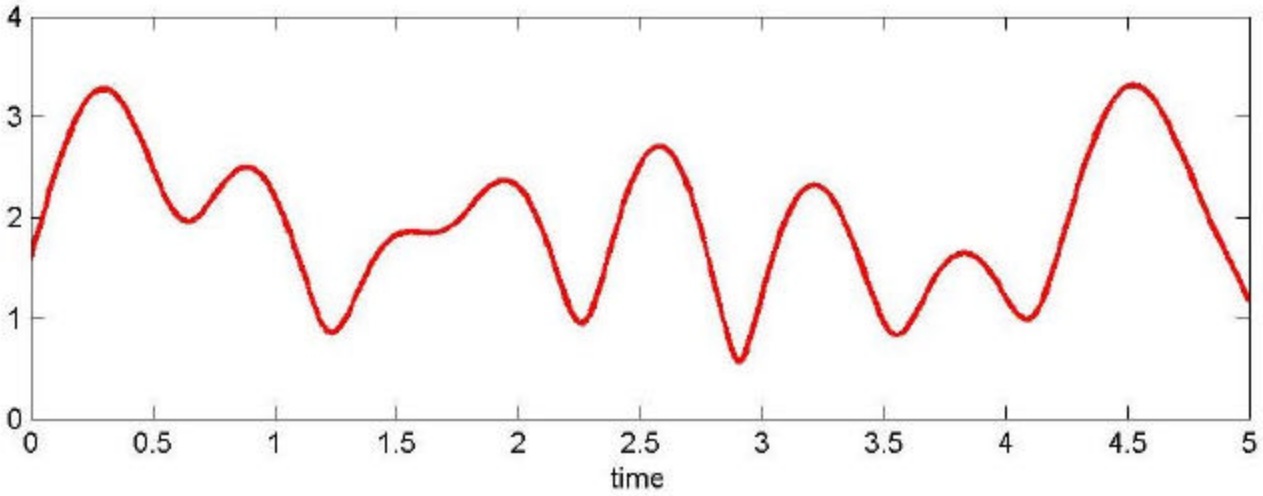
- Functions of Physical Layer
- **Theoretical Basis for Physical Layer Operations**
  - Analog v.s. Digital
  - Bandwidth
  - Modualation
- Guided Transmission Media
- Terrestrial Wireless Transmission
- Communication Satellite

THEORY: TIME DOMAIN SIGNALS

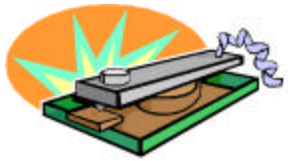
- **Analog Message**
  - A message that is defined on a continuum, i.e., the signal can take infinitely many values.
  - Waveform fidelity is important.
  - Example: human speech.



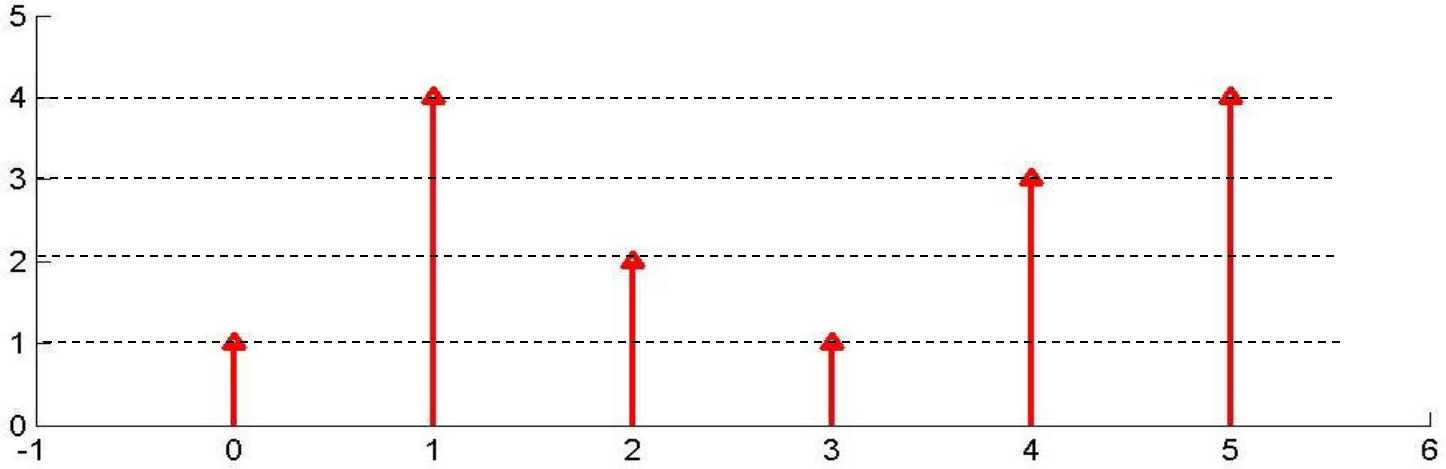
- **Analog Signal**
  - The electrical signal used to represent analog message



THEORY: TIME DOMAIN SIGNALS

- **Digital Message**
  - A Message that has finitely many possible values.
  - Example: traffic lights  Telegraph 

- **Digital Signal**
  - The electrical signal used to represent digital message.



THEORY: ANALOG V.S. DIGITAL

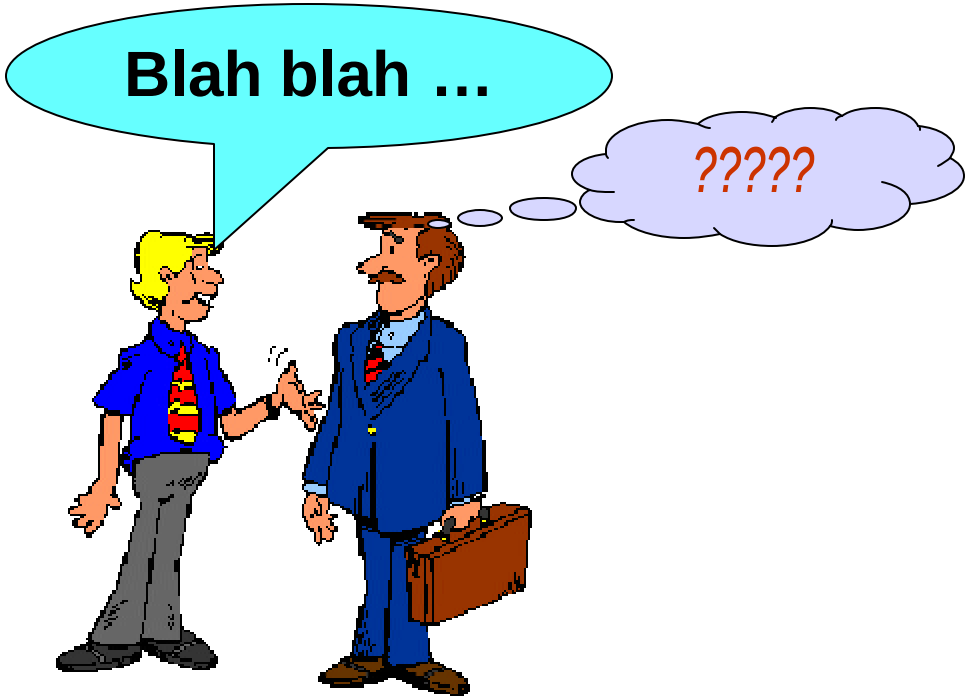
- **Usually, digital signals are used for digital message**
  - E.g., computer bit on Ethernet wire
- **Also, analog signals are used for analog message**
  - E.g., voice over telephone wire
- **However, one can use analog signal to carry digital message**
  - E.g., computer bits over telephone wire via a Modem
- **Also, analog message can be converted to digital message, then digitalized message can be carried by digital signal.**
  - Through sampling and quantization
  - E.g., audio in a MP3 player
- **Also, digital message can be converted to analog message, then the converted message can be carried by analog signal.**
  - Through interpolation
  - E.g., photo from digital camera displayed on a regular TV

THEORY: ANALOG V.S. DIGITAL

- **Analog Communication System**
  - Transfers information from an analog message source to a destination (sink)
    - Analog message is transmitted in channel
    - Recover analogy message at receiver
  - Objective: keep the fidelity of the analog waveform.
  - Example: AM/FM radio
- **Digital Communication System**
  - Transfers information from a digital message source to a destination (sink)
    - Digital message is transmitted in channel.
    - Recover digital message at receiver
  - Objective: Recover the digital message at the receiver.
    - Waveform fidelity is no longer important.
  - Example: Computer Network.

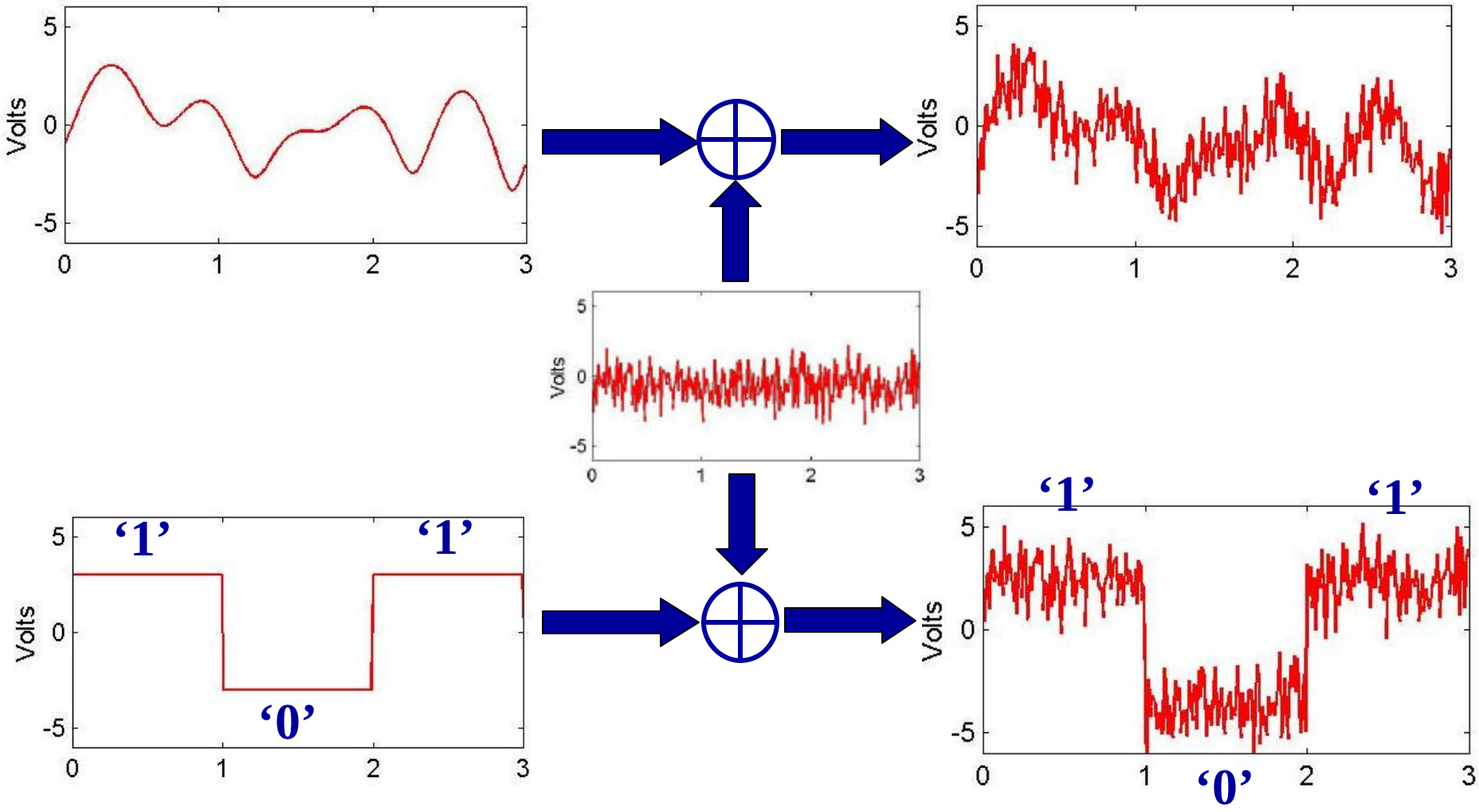
THEORY: ANALOG V.S. DIGITAL

- **When two people are talking with each other face by face, is this an analog communication system or digital communication System? What if they talk over cell phone?**



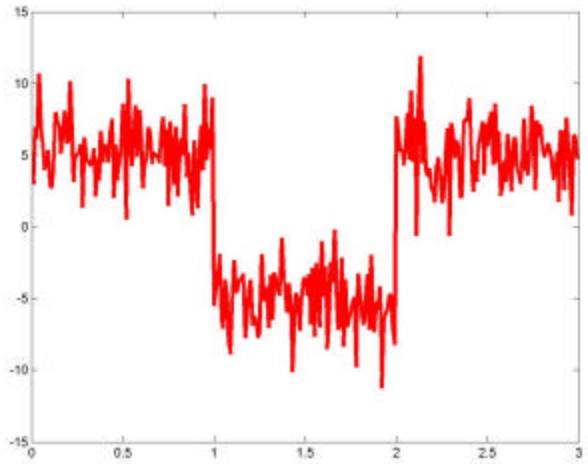
What matters is what kind of message is transmitted in channel.

THEORY: WHY DIGITAL?

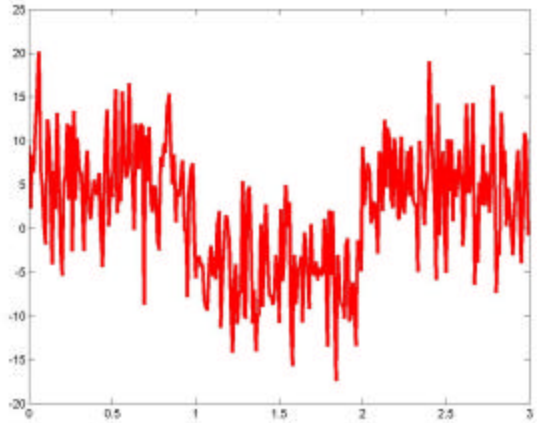


THEORY: SIGNAL TO NOISE RATIO

- **Signal to noise ratio (SNR): The ratio of the signal power to the noise power at the receiver.**
  - SNR = S/N, with S being the signal power, and N being the noise power observed by the receiver.
  - High SNR ➔ Signal is strong, and noise is weak ➔ Better communication quality.
  - Improve SNR ➔ Improve Tx power ➔ More power consumption



High SNR



Low SNR

THEORY: SIGNAL TO NOISE RATIO

- **SNR is usually defined in the unit of dB (decibel)**
  - Linear value: SNR = S/N (no unit)
  - dB value: SNR(dB) = $10\log_{10}(S/N)$ dB
  - S/N = 2 ➔ SNR (dB) = $10\log_{10} 2$ = 3dB

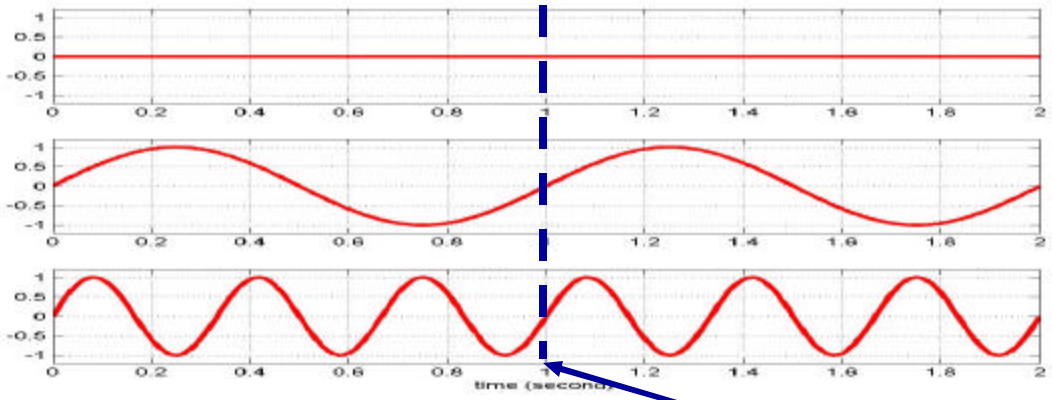
- **Example:**
  - The proper operation of the 1st generation cell phone system (Analog) needs to maintain an SNR of approximately 18dB.
  - The proper operation of the 2nd generation cell phone system (Digital) needs to maintain an SNR of approximately 14dB.
  - Low SNR requirement ➔ Low Tx power ➔ Long battery life.
  - Even with less SNR requirement, the communication quality of the 2nd generation system is much better than that of the 1st generation system

THEORY: WHY DIGITAL?

- **Better Noise Immunity**
  - Better Communication Quality.
- **Ease of Encryption**
  - Better Privacy.
- **Error might be corrected by channel coding.**
- **Voice, video, and data can be transmitted over a common system.**
  - E.g.: Cell phones.
- **Digital message can be transmitted over analog system.**
  - E.g.: HDTV and analog TV programs are transmitted through the same cable network.
  - E.g.: Dial-up modem

THEORY: CONCEPT OF FREQUENCY

- **Frequency is the measurement of the number of times that a repeated event occurred in a unit time (1 second).**
  - Frequency measures how fast a signal can change within a unit period of time (or the measurement of the rate of change).
  - High frequency ➔ the signal changes fast
- **In signal processing, frequency is defined based on sinusoid signal → Number of oscillation per second.**
  - Each sinusoid signal is uniquely associated with a single frequency based on its rate of change.



f = 0 Hz

f = 1 Hz

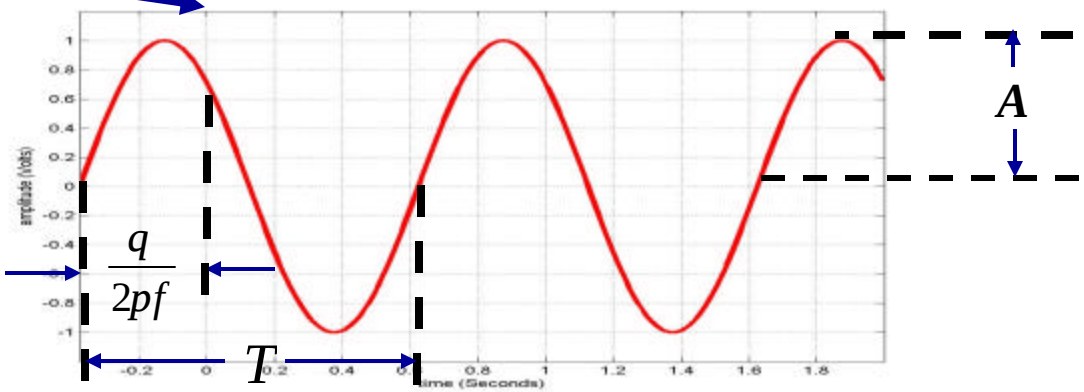
f = 3 Hz

THEORY: CONCEPT OF FREQUENCY

- **Sinusoid Signal**

$$s(t) = A\sin(2\pi ft + q)$$

  - f : frequency. in the unit of Hz (1/second)
  - A : amplitude. maximum strength in signal, usually in unit of volts
  - q : phase. relative position in time
  - t : time. in the unit of second
  - T : period. Time for one cycle or one repetition, $T=1/f$.
  - l : Wavelength. Distance of the signal waveform propagated in one period T.
    - $l = vT$, where v is the speed of the signal. In vacuum, it is the speed of light.

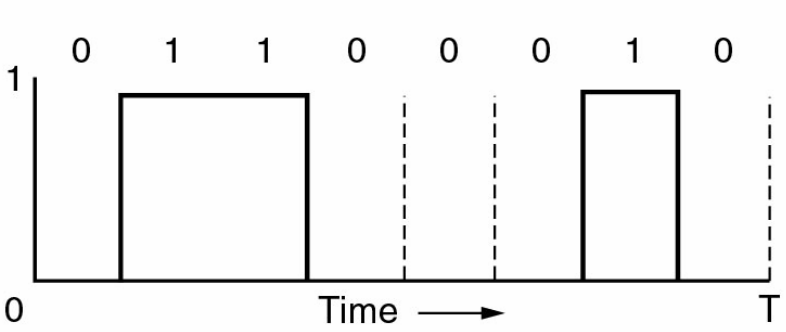


THEORY: FREQUENCY DOMAIN SIGNAL

- **Periodic signal can be written as the summation of a series of sinusoids (Fourier Series)**

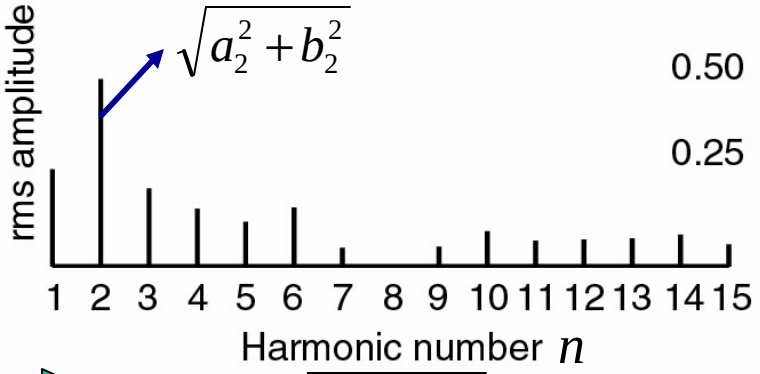
$$s(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2pnf_n t) + \sum_{n=1}^{\infty} b_n \cos(2pnf_n t)$$

  - Each sinusoid component has a unique frequency
  - Each sinusoid component is called a "harmonic" of the original signal.
  - There is a one to one relationship between a signal and its Fourier series.
  - With Fourier series, we can represent the signal in the "Frequency Domain"



$s(t)$

Time Domain



$\sqrt{a_n^2 + b_n^2}$

Frequency Domain

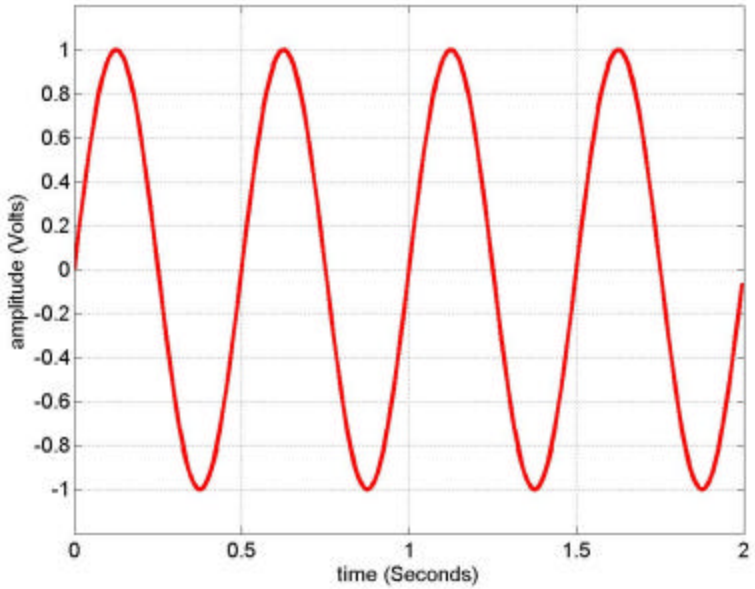
THEORY: FREQUENCY DOMAIN SIGNAL

- **Similarly, we can represent general signals in frequency domain with Fourier transform.**

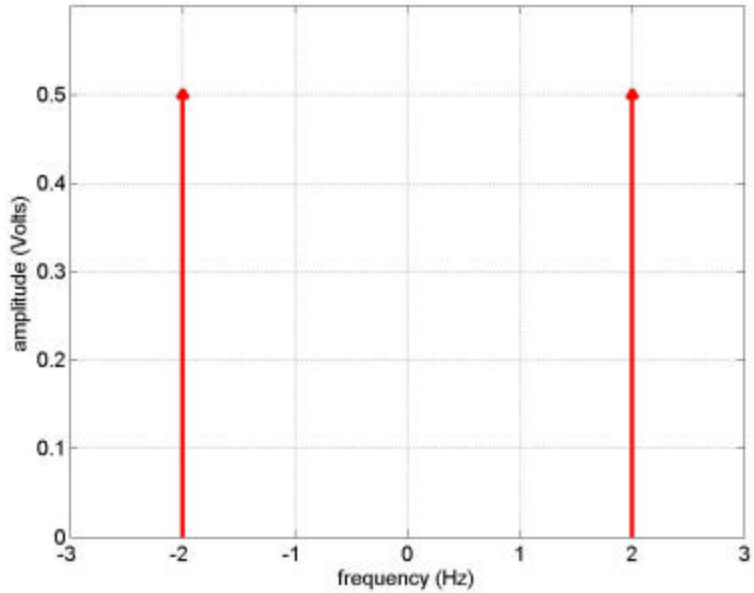
$$s(t) = \sin(2pf_0 t)$$

Fourier Transform

$$S(f) = \frac{i}{2}[d(f + f_0) - d(f - f_0)]$$



Time Domain



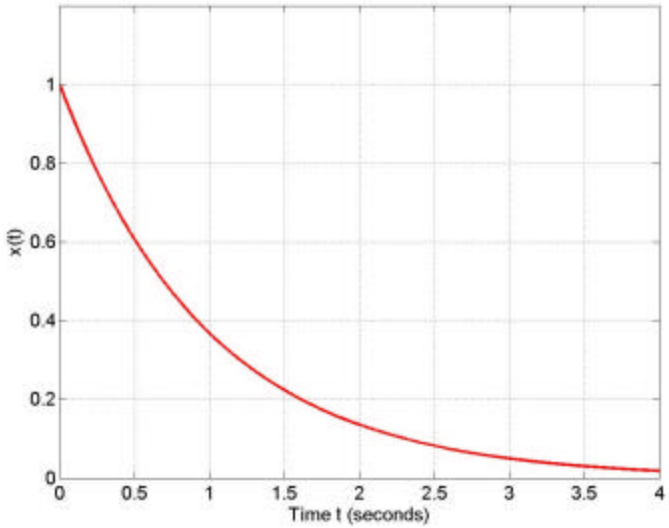
Frequency Domain

THEORY: FREQUENCY DOMAIN SIGNAL

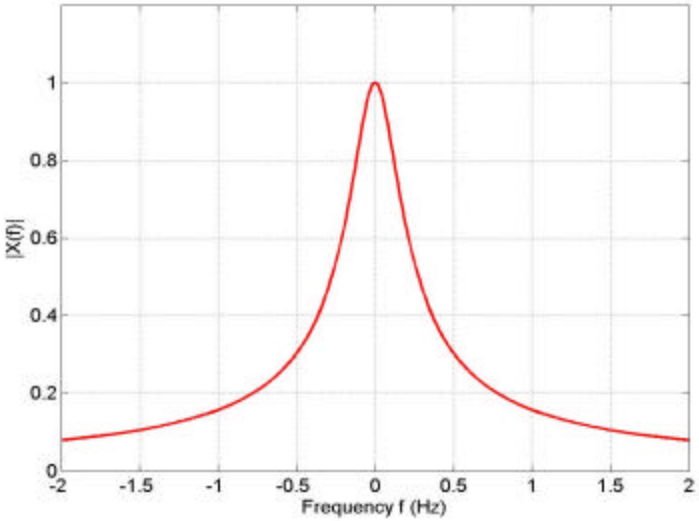
$$s(t) = e^{-t}, \quad t \geq 0$$

Fourier Transform

$$S(f) = \frac{1}{1 + j2pf}$$



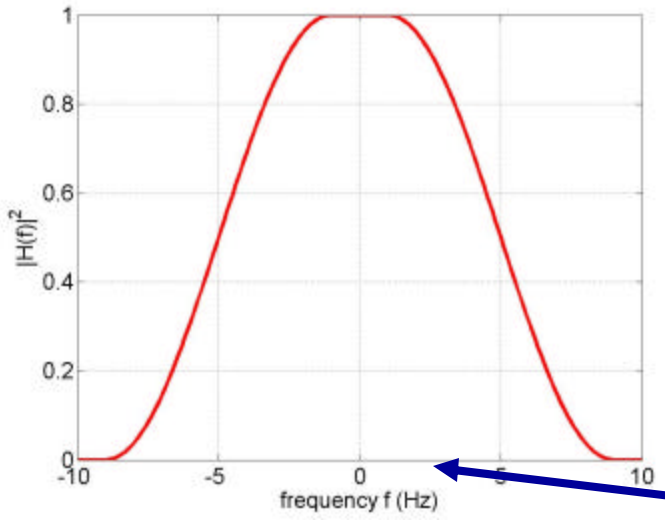
Time Domain



Frequency Domain

THEORY: BANDWIDTH

- **Signal will lose power during Tx in the communication system.**
- **For practical communication system, signal components at different frequency will experience different attenuation (power loss).**
- **The range of frequencies transmitted without being strongly attenuated is called bandwidth.**



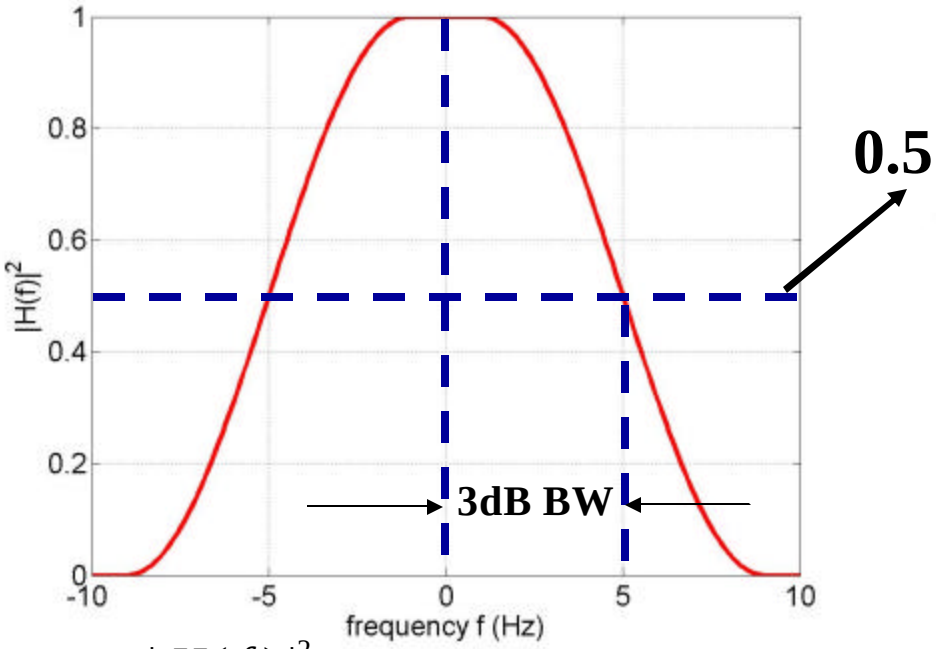
The frequency domain representation of a typical communication channel

frequency domain

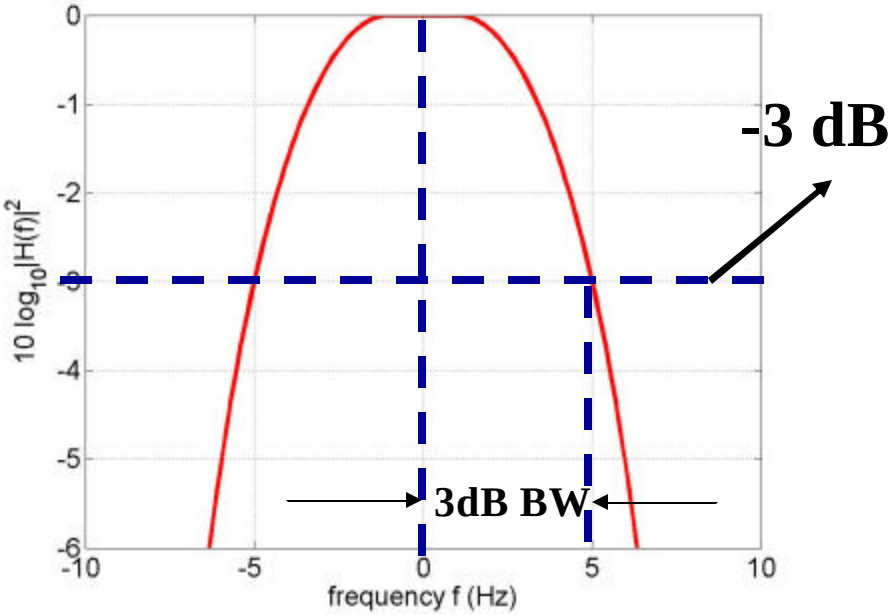
THEORY: BANDWIDTH (BW)

- **3dB Bandwidth (Half Power Bandwidth)**
  - The bandwidth is defined as from 0 Hz to the frequency at which half power get through.
  - In log domain, half power (0.5) corresponding to 3dB attenuation.

$10\log_{10}(1) = 0\text{dB}$ $10\log_{10}(0.5) = -3\text{dB}$



$|H(f)|^2$

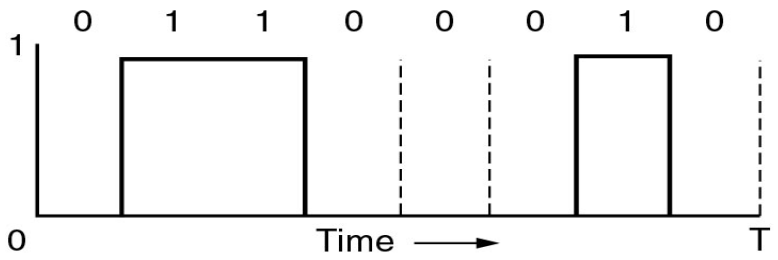


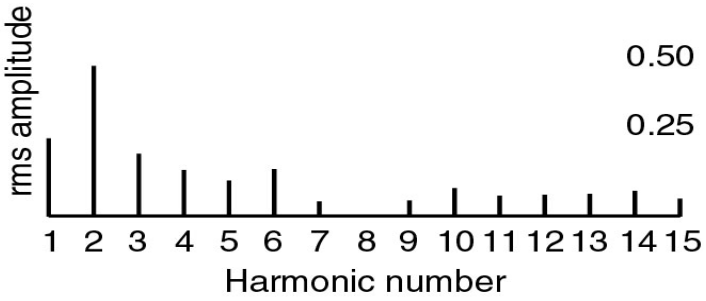
$10\log_{10}|H(f)|^2$

THEORY: BANDWIDTH

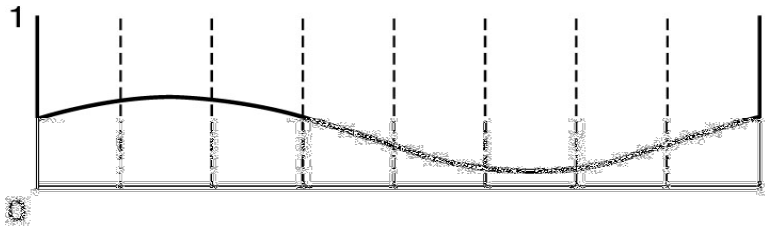
- **Signal Bandwidth:**
  - the bandwidth of the Tx signal.
  - Usually proportional to the data rates.
    - Data rate represents how many bits is carried out by the signal per second.
    - It is proportional to how fast the signal will change per second.
- **Channel Bandwidth:**
  - the bandwidth of the Tx medium.
  - Channel Bandwidth is a physical property of the Tx medium
    - Depends on the material, thickness, length and other factors of the medium
  - Sometimes a filter is introduced to limit the channel BW available to a user.
    - E.g., telephone line has a BW of approximately 1MHz (physical bandwidth), but telephone company adds a filter to restrict the BW to 3100 Hz (artificial bandwidth).
- **When the signal bandwidth is larger than the channel bandwidth, the signal will be significantly distorted.**

THEORY: BANDWIDTH LIMITED SIGNAL

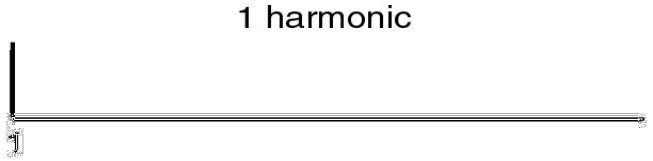


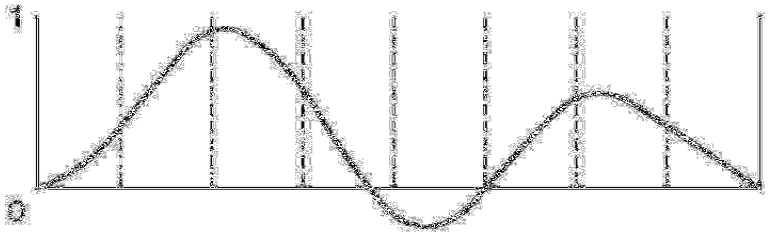


(a)

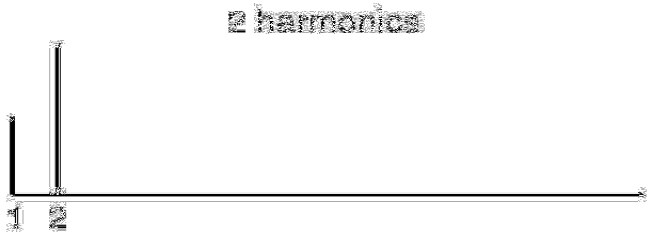


(b)

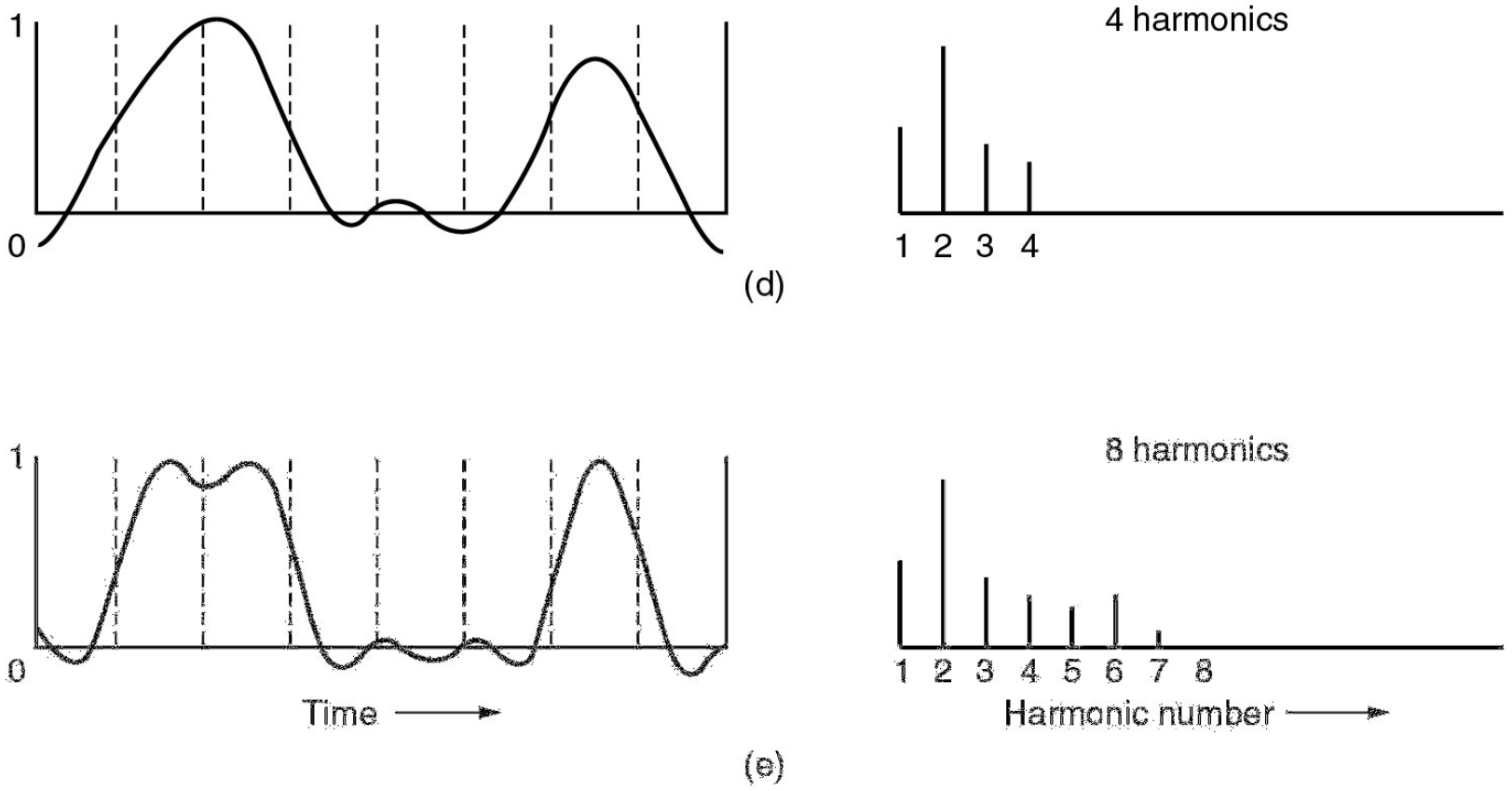




(c)



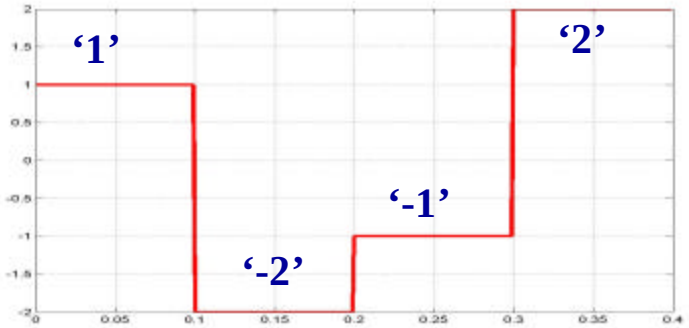
THEORY: BANDWIDTH LIMITED SIGNAL (CONT'D)



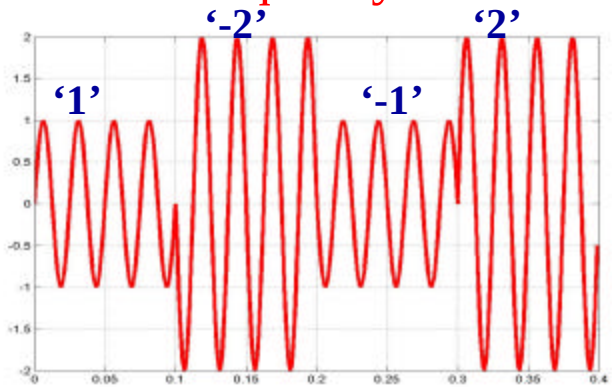
THEORY: MODULATION

- **Why modulation?**
  - Transfer the original binary information to an electrical waveform that is suitable for transmission in the physical medium.
    - Limited BW
    - High frequency ➔ Further propagation distance
- **Amplitude Modulation, Frequency Modulation, Phase Modulation**
  - The sinusoid is called the information carrier
  - The frequency of the sinusoid is called carrier frequency.

AM



Original Signal



Amplitude Modulated Signal

THEORY: MODULATION

A binary signal

Amplitude modulation

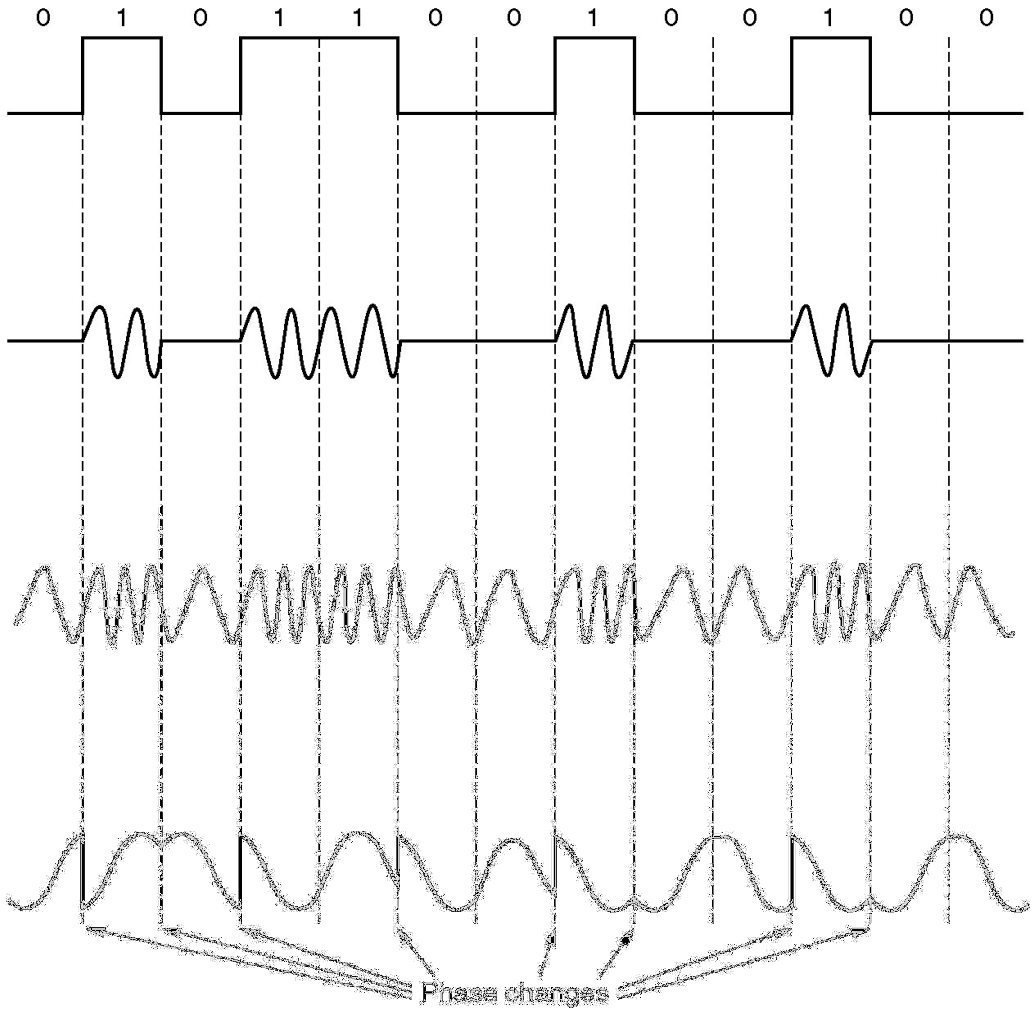
E.g., A_o="0", A_1="1"

Frequency modulation

E.g., f_o="0", f_1="1"

Phase modulation

E.g., 0°="0", 180°="1"



THEORY: MODULATION

- **Each modulation symbol can contain several bits of information**
  - The number of symbols available in modulation is called the modulation alphabet size.
  - E.g. Four amplitudes, {-2v, -1v, 1v, 2v}, are used in modulation, then each amplitude (symbol) represents two-bit of information.
  - # of bits/symbol = $\log_2$(modulation alphabet size)
- **Baud**
  - The number of modulation symbols per second.
    - Inverse of symbol period.
- **Bit Rate**
  - The number of bits being transmitted per second
  - Bit Rate = Baud x (# of bits/symbol)
- **Bit Rate (bps) can be improved by increasing the modulation alphabet size.**

THEORY: MODULATION

- **Fundamental trade-off between data rate and system performance.**

  Given bandwidth and SNR

  – Larger modulation alphabet size ➔ more bits/symbol ➔ lager data rate ☺

  – Lager modulation alphabet size ➔ receiver needs to discriminate among more symbols ➔ larger probability of making errors ➔ performance degradation ☹

THEORY: BANDWIDTH AND MAXIMUM DATA RATE

- **For a system with bandwidth *H* Hz**
  - If there is no noise present in the system, and the signal is represented by *V* discrete levels (V: modulation alphabet size), the maximum data rate that can be achieved by the system is

$$2H \log_2 V \text{ bps}$$

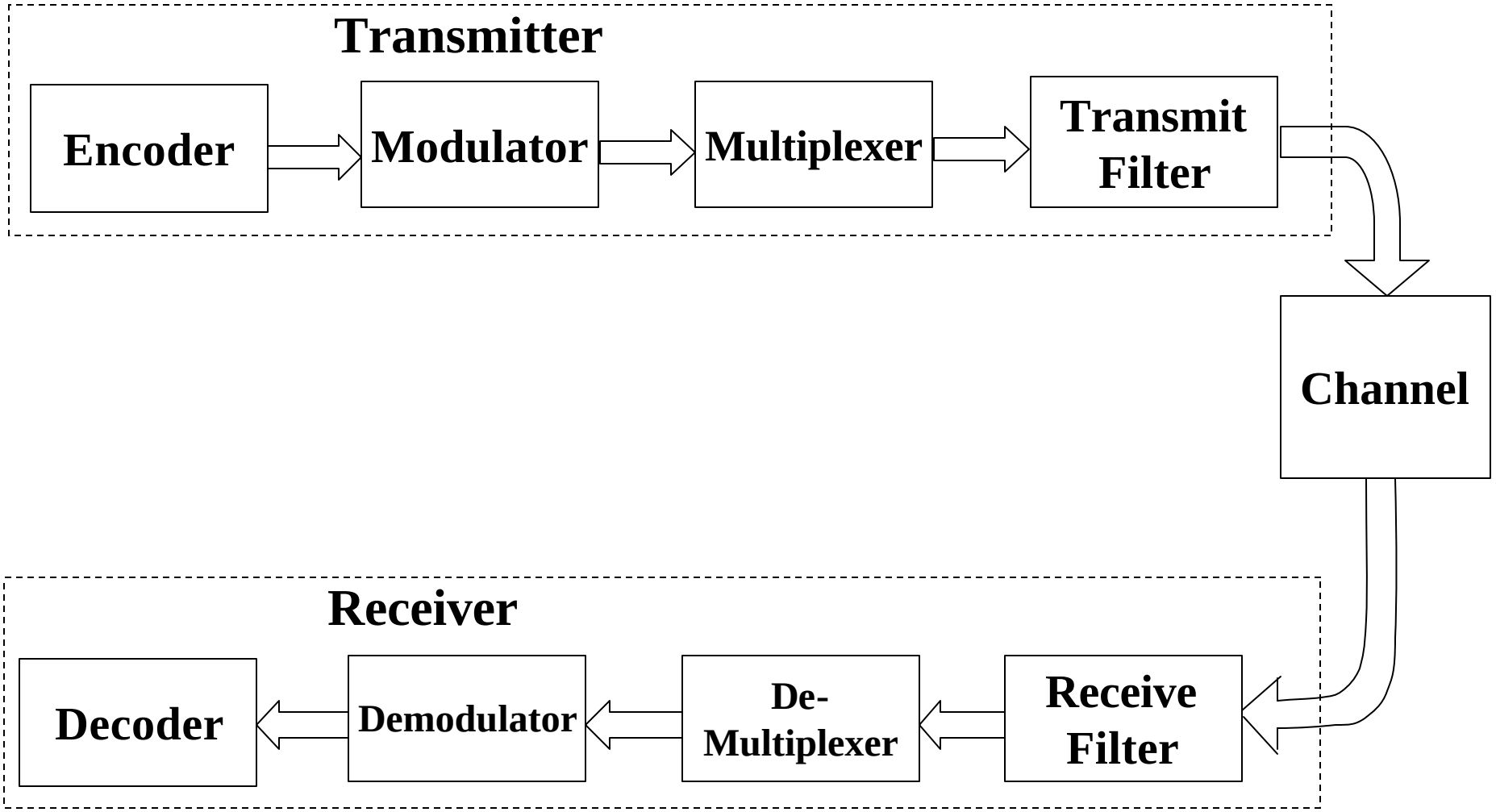
  - If there is noise present in the system and the signal to noise ratio is S/N, then the maximum data rate (capacity) that can be achieved by the system is

$$H \log_2 (1+S/N) \text{ bps}$$

THEORY: SIMPLEX AND DUPLEX

- **Simplex**
  - Tx occurs only in one direction
  - E.g. TV
- **Half Duplex**
  - Tx can occur in either direction, but only one way at a time.
  - E.g. Walki-Talki (police radio)
- **Full Duplex**
  - Tx can occur at both direction simultaneously
  - E.g. Telephone.

THEORY: SIMPLIFIED PHYSICAL LAYER DIAGRAM



OUTLINE

- Functions of Physical Layer
- Theoretical Basis for Physical Layer Operations
- **Guided Transmission Media**
- Terrestrial Wireless Transmission
- Communication Satellite

GUIDED MEDIA: CLASSIFICATIONS

- **Classifications of guided Tx media**
  - Removable Media
    - Magnetic tape
    - Floppy disk, zip drive
    - DVD
    - Flash drive
  - Twisted Pair
    - Phone line
  - Coaxial Cable
    - Cable TV
  - Fiber Optics
    - Network backbone

GUIDED MEDIA: REMOVABLE MEDIA



- **Communication process for removable media**
  - Write data in the removable media at the source computer
  - Physically move the removable media to destination computer
  - Read out the data from the removable media at destination computer.
- **One of the most common ways to transport data from one computer to another.**
- **Disadvantage: not suitable for delay sensitive or real time services.**
  - Delay is measured in minutes, hours, and even days.
  - Most service requires a delay in the order of milliseconds
- **Advantage: Good bandwidth efficiency, cost effective.**
  - Can achieve a very high data rate at a relatively low cost.

GUIDED MEDIA: TWISTED PAIR

- **Twisted pair is one of the oldest yet still most common transmission media.**
  - A twisted pair consists of two insulated copper wire twisted together.
- **Why twisted?**
  - Two parallel wires makes a fine antenna.
  - Emitting energy of the Tx signal.
  - Interfering with other wires (cross talk).



(a)

Category 3
- Up to 16 MHz
- Twist length: 7.5cm to 10cm
- Voice grade found in most buildings



(b)

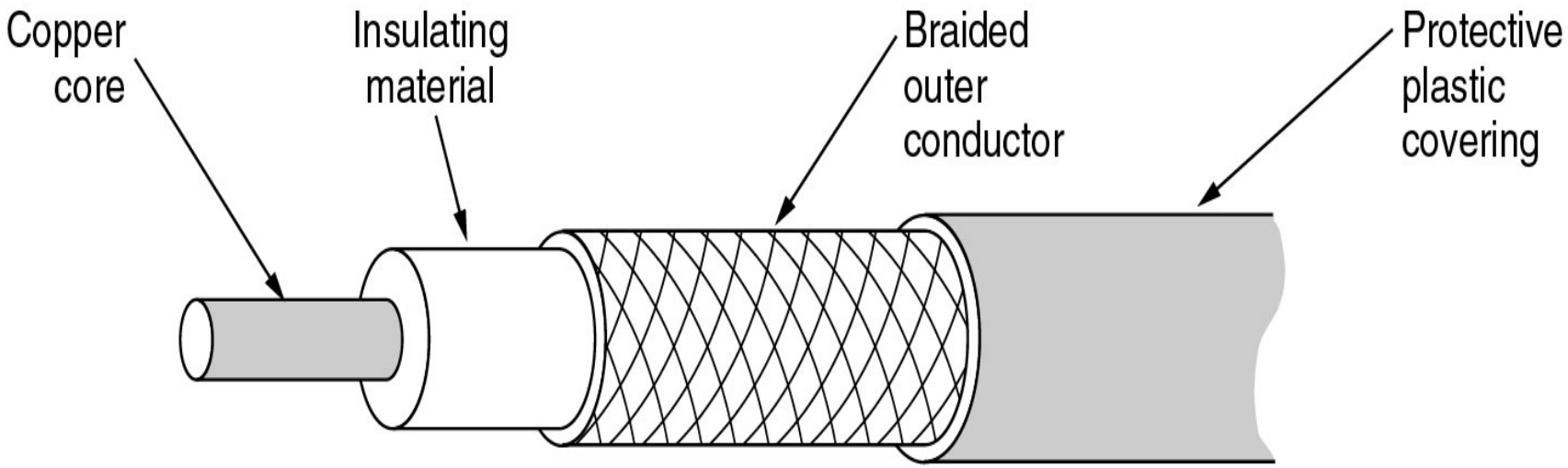
Category 5
- Up to 100 MHz
- Twist length: 0.6cm to 0.85cm
- pre-installed in new buildings (after 1988)

GUIDED MEDIA: TWISTED PAIR

- **When many twisted pairs run in parallel, they are usually bundled together and put in a protective shield.**
  - If each pair is not twisted, they will generate interference for each other.
- **Characteristics of twisted pair**
  - Can be used for transmitting either digital signal or analog signal
    - Telephone: analog signal
    - Local area network: digital signal
  - Bandwidth varies from 1MHz ~ 600 MHz
  - For a few kilometers, data rates can be achieved in the order of several megabits/sec
  - Can run several kilometers without amplification
  - Easy to tap.

GUIDED MEDIA: COAXIAL CABLE

- **Coaxial cable (often referred to as co-ax)**
  - Has better shielding than twisted pairs
  - Compared to twisted pair, it span longer distance at higher speed.
  - Has a bandwidth of close to 1GHz.

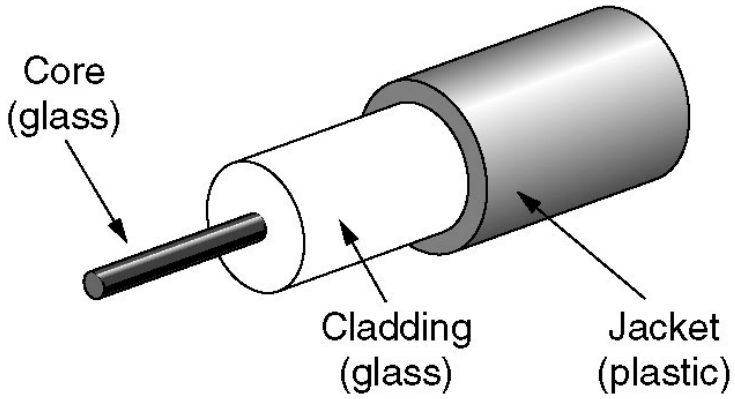


GUIDED MEDIA: COAXIAL CABLE

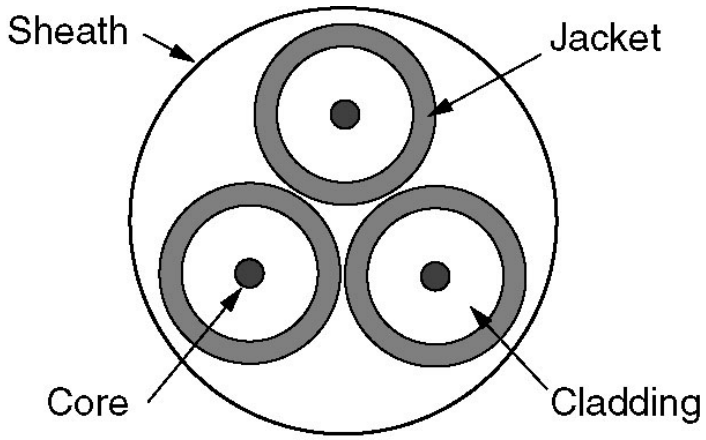
- **Two transmission types**
  - Analog
    - Amplifiers needed every few km
      - Closer for higher frequencies
    - Up to 500 MHz
  - Digital
    - Repeater every 1 km
      - Closer for higher data rates
- **Applications**
  - TV distribution
  - Long distance telephone transmission
  - Short-range high-speed device connections
  - Local area networks

GUIDED MEDIA: FIBER OPTICS

- **Fiber Optic Cable**
  - Glass core surrounded by a glass cladding
  - Glass core is made from super transparent glass
  - Glass cladding has a lower index of refraction than the core
    - Lights are kept in the core



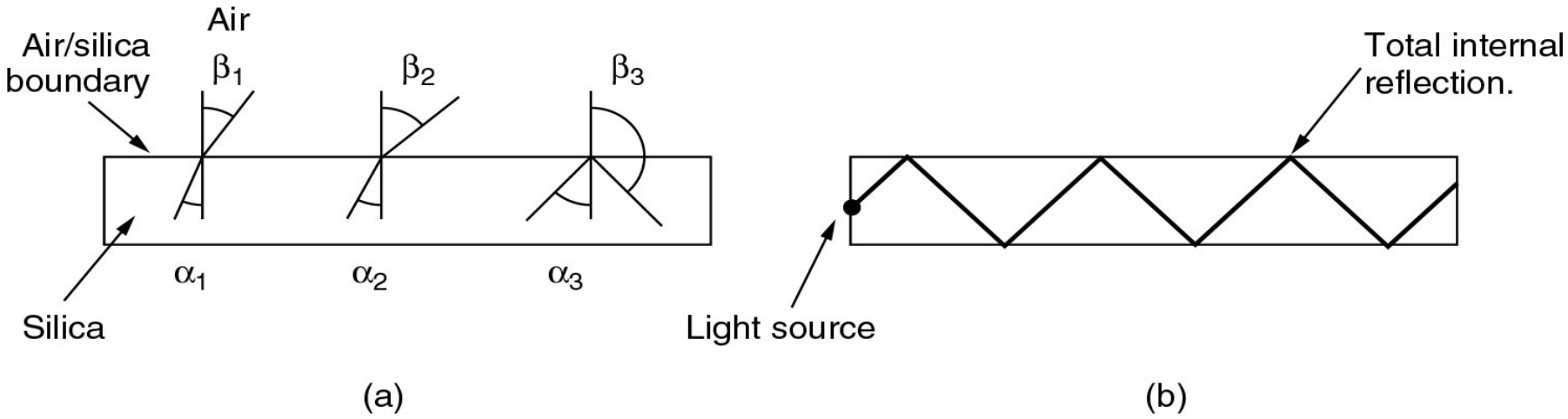
(a)



(b)

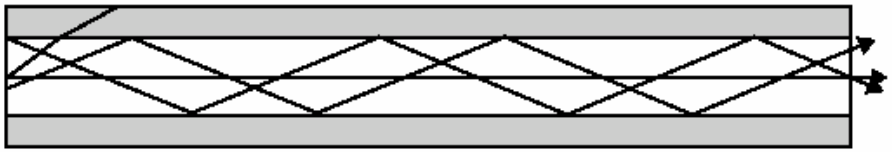
GUIDED MEDIA: FIBER OPTICS

- **Why light will be kept in the glass core?**
  - Glass core and glass padding are two mediums with different indices of refraction
  - Light is inside in the medium with larger index of refraction (glass core) than that of the medium outside (glass padding, or air).
  - There is a critical angle determined by the index of refraction
    - If the angle of incidence of the light is above the critical angle, then the light is completely reflected back to the glass core.



GUIDED MEDIA: FIBER OPTICS TX MODE

- **Multimode Fiber**
  - Multiple light rays with different incident angle can coexist in a single fiber
  - Each ray is said to have a different mode, and the fiber is called multimode fiber.



- **Single-mode Fiber**
  - If the fiber's diameter is reduced to a few wavelengths of light, the light can propagate only in straight line without bouncing ➔ single mode fiber
  - More expensive, more widely used for long distance
  - Current fiber can achieve Tx data at 50Gbps for 100 km without amplification.

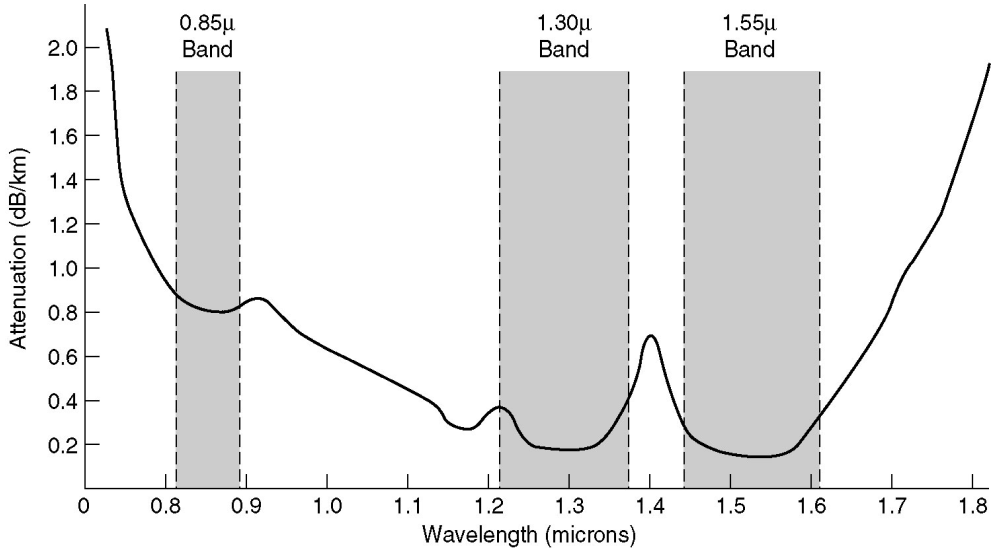


GUIDED MEDIA: FIBER OPTICS - HOW DOES IT WORK?

- **Three key components**
  - Light source
    - Generate pulse of lights based on input electrical signal.
    - A pulse of light indicates a ‘1’ bit
    - Absence of light indicates a ‘0’ bit
  - Transmission medium
    - Fiber optic cable
  - Light detector
    - When hit by light, generate an electrical pulse
- **Attaching a light source to one end of a cable, and a detector to the other end, we have a unidirectional communication system**
  - Electrical signal ➔ light ➔ electrical signal

GUIDED MEDIA: TX OF LIGHT THROUGH FIBER

- **Fiber optic class is made of super transparent glass**
  - If the water in ocean is replaced by the glass used in fiber, the sea bed will be clearly visible from the surface.
- **Glass will attenuate the light Tx through it**
  - Attenuation depends on light wavelength.
  - Attenuation in decibels $10 \log_{10} \frac{\text{Tx Power}}{\text{Rx Power}} (\text{dB})$
  - A factor of two loss gives an attenuation of $10 \log_{10} 2 = 3(\text{dB})$



Three bands are used

- each has BW of 25,000 to 30,000 GHz
- 1 micron = 10^{-6}m
- visible light has wavelength of 0.4 to 0.7 microns

GUIDED MEDIA: FIBER OPTICS

- **Connections of fibers**
  - Fiber can terminate in connectors and plugged into fiber sockets
    - Connector lose 10%-20% of light
    - Easy to tap
  - Two fibers can be spliced mechanically
    - Lay two carefully-cut ends next to each other in a special sleeve
    - Clamp them together
    - 10% light loss if done by trained personnel
  - Two fibers can be fused (melted) together
    - Minimum light loss
- **All three kinds of connections will result in reflection at the point of splice**
  - Reflected light will interfere with original light.

GUIDED MEDIA: FIBER OPTICS

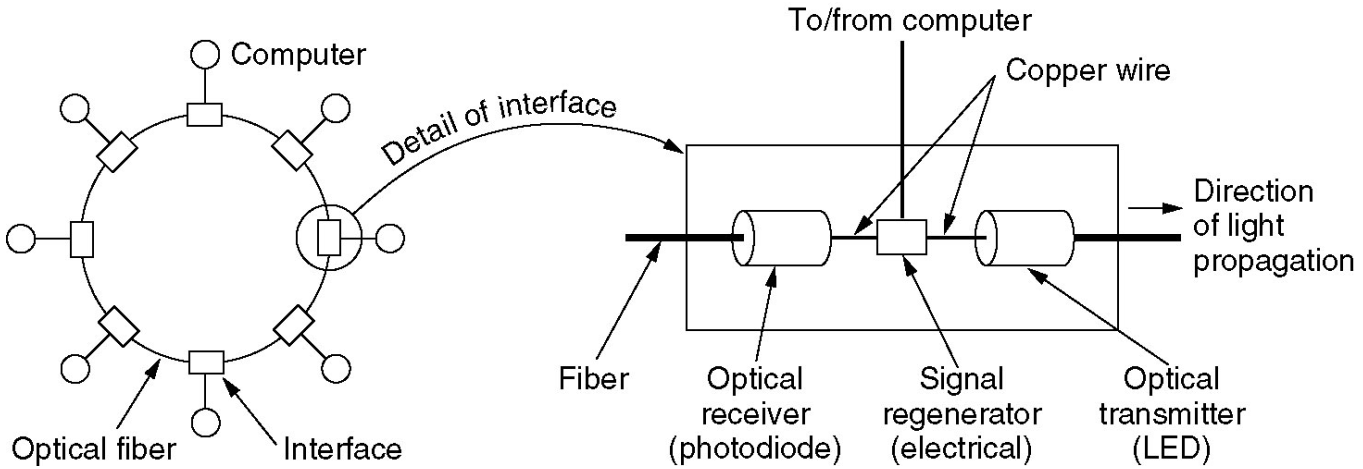
- **Light Sources**
  - LED (Light Emitting Diode)
  - Semiconductor Laser

Item	LED	Semiconductor laser
Data rate	Low	High
Fiber type	Multimode	Multimode or single mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

- **Light Detectors**
  - Photodiode ➔ gives off electrical pulse when hit by light
  - Typical response time 1ns ➔ Space between two light pulses cannot be less than 1ns ➔ data rates up to 1Gbps

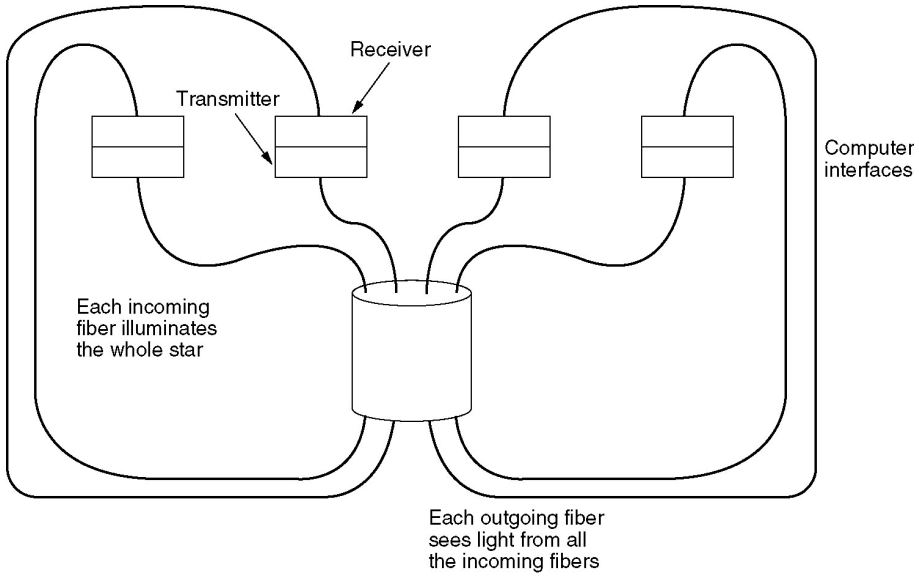
GUIDED MEDIA: FIBER OPTIC NETWORKS

- **Fiber optics can be used in both LANs and long-haul transmission.**
- **Two types of interface**
  - Passive interface
    - Two taps fused onto the main fiber, one with light source and the other one with photodiode.
    - A broken LED or photodiode will not break the network.
  - Active repeater
    - Incoming light is converted to electrical signal
    - Electrical is amplified if necessary
    - The amplified signal is converted to light and sent out.
    - Pure optical regenerator is available now.
    - If an active repeater fails, the ring goes down.



GUIDED MEDIA: FIBER OPTIC NETWORKS

- **Passive star topology**
  - Fibers from transmitters are fused onto a silica cylinder
  - Fibers fused to the other end of the cylinder are fused to receivers.
  - If one interface emits light, all the receivers will be illuminated ➔ broadcasting
  - Strength of lights are divided among receivers ➔ The number of nodes is limited by the sensitivity of the photodiode.



GUIDED MEDIA: FIBER OPTICS V.S. COPPER WIRE

- **Advantages**
  - Fiber has much higher bandwidth
  - Low attenuation➔ repeater only needed every 50km (copper requires repeaters every 5km.
  - Immune to electromagnetic interference
  - Thin and lightweight
  - Does not leak light and difficult to tap ➔ high security

- **Disadvantages**
  - Can be easily damaged by being bent too much
  - Difficult to tap
  - Fiber interface cost most than electrical interfaces

OUTLINE

- Functions of Physical Layer
- Theoretical Basis for Physical Layer Operations
- Guided Transmission Media
- **Terrestrial Wireless Transmission**
- Communication Satellite

WIRELESS TRANSMISSION

- **Electromagnetic Waves**
  - The movement of electrons creates electromagnetic waves
  - Electromagnetic waves can propagate through space
    - Vacuum
    - Air
    - Water
- **The propagation of electromagnetic wave through space builds the foundation of wireless communication.**



WIRELESS TRANSMISSION

- **Frequency, Speed, Wavelength**
  - Frequency f
    - Number of Oscillations Per Second of the electromagnetic wave.
  - Speed c
    - In vacuum, electromagnetic waves travel at the speed of light:

      $$c = 3\times10^{8} \text{ m/sec}$$

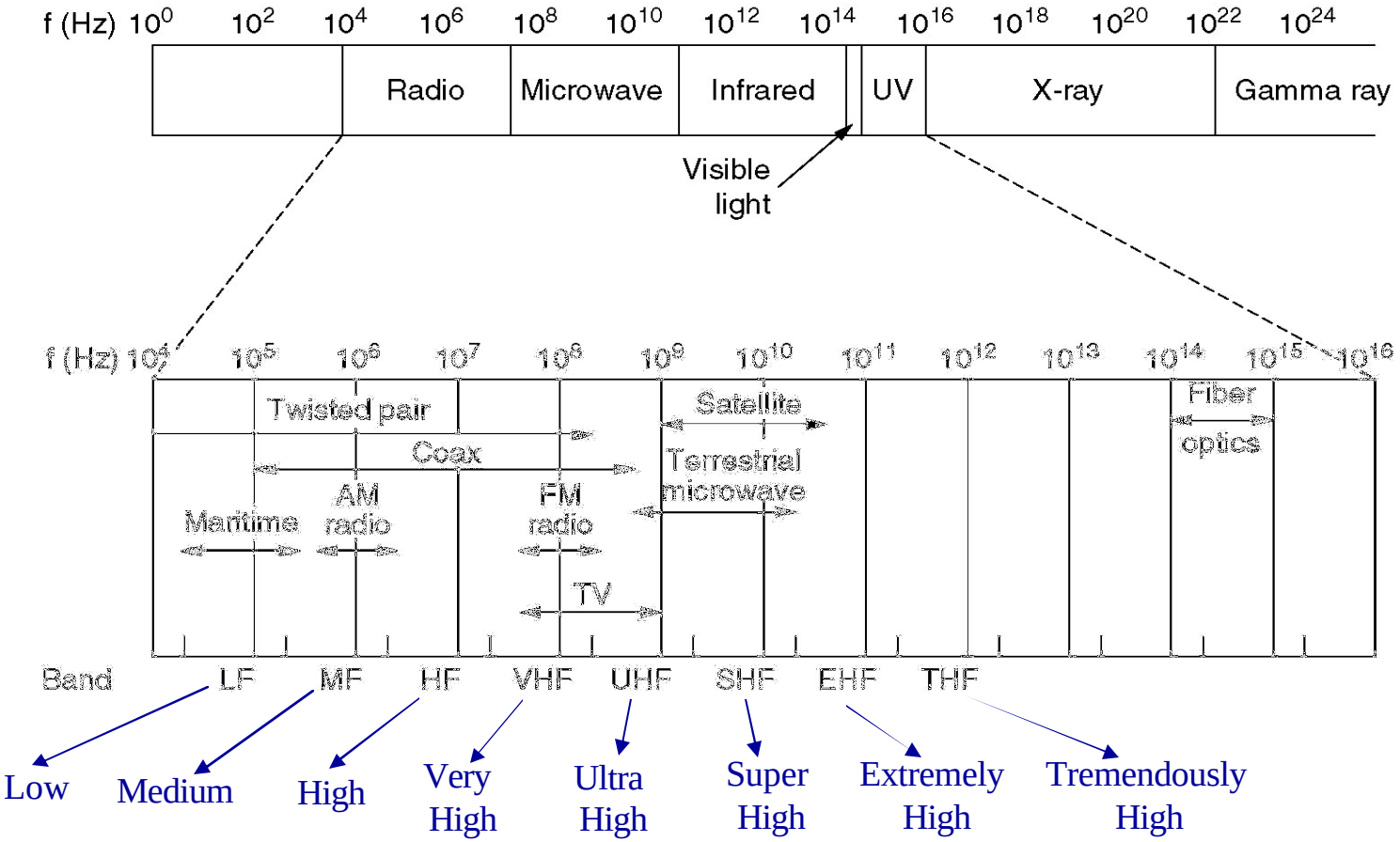
    - In copper or fiber, the speed slows to about 2/3 of c.
  - Wavelength λ
    - The distance corresponding to one period of the waveform.
- **Relationship between frequency, propagation speed, wavelength**

  $$c = \lambda f$$

  - $f \downarrow \Rightarrow \lambda \uparrow$, $f \uparrow \Rightarrow \lambda \downarrow$

WIRELESS TRANSMISSION

- **Electromagnetic Spectrum**

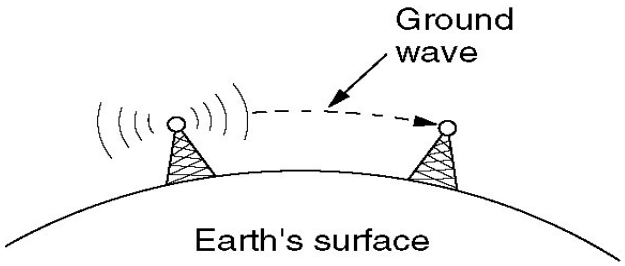


WIRELESS: RADIO TRANSMISSION

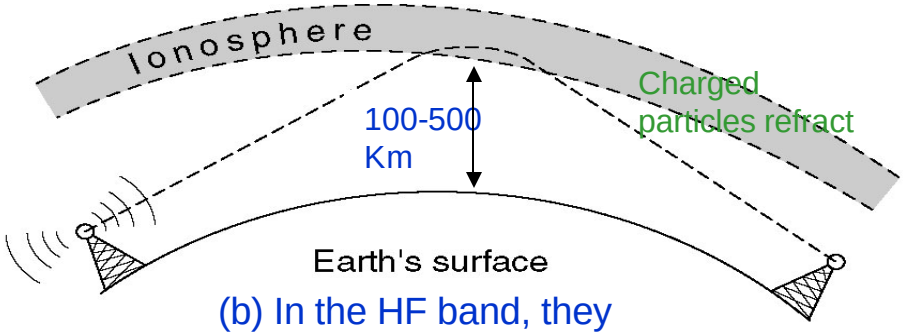
- **Characteristics**
  - Approximately from 10 KHz ~ 1 GHz
  - Easy to generate, travel long distance, penetrate buildings
  - Omni-directional: travel to all directions from the source.
  - Subject interference from motors, electrical equipment, and other users operating at the same frequency
    - The radio transmission is tightly licensed by government.
- **Frequency dependent properties**
  - At low frequencies,
    - Pass through obstacles well
    - Power falls off sharply with propagation distance.
  - At high frequencies
    - Travel in straight line
    - Bounce off obstacles
    - Like light
- **Applications**
  - AM radio, FM radio, Amateur radio, TV broadcast

WIRELESS: RADIO TRANSMISSION

- **In VLF, LF, MF bands, radio wave propagates follow the ground**
  - Pass through buildings easily
  - Low bandwidth
- **In HF and VHF bands, radio waves can be bounced off ionosphere**
  - Ionosphere: a layer of charged particles circling the earth at a height of 100 to 500km.



(a) In the VLF, LF, and MF bands, radio waves follow the curvature of the earth



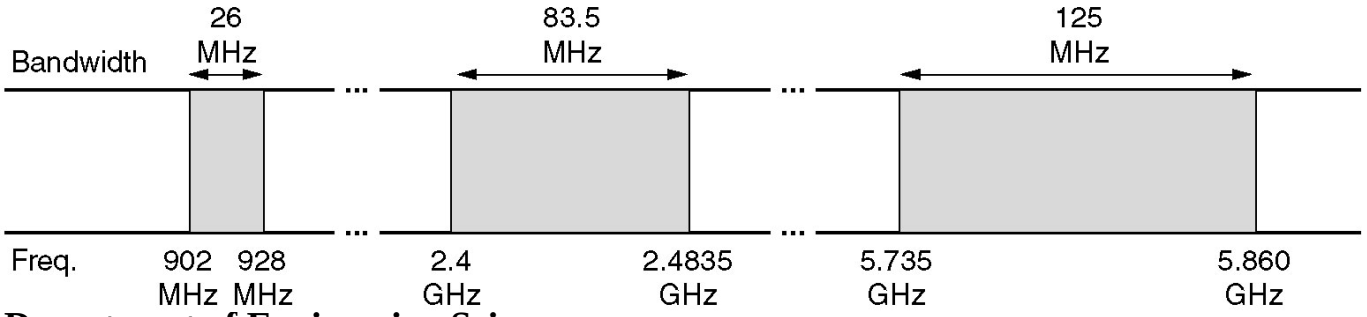
(b) In the HF band, they bounce off the ionosphere

WIRELESS: MICROWAVE TRANSMISSION

- **Characteristics**
  - Approximately 1GHz ~ 300GHz (wave length: 300 mm ~ 1 mm)
  - Travel in nearly straight lines
  - Can be highly directional
    - Focus all energy into a small beam gives a higher SNR
    - Tx and Rx antennas must be accurately aligned with each other.
  - Inexpensive
    - Doesn't need to buy right of way
    - Putting up two towers is much cheaper than laying out 50km fibers.
  - Doesn't pass through buildings well.
- **Applications**
  - Long distance telephone transmission
    - MCI use towers tens of kilometers apart. (The curvy shape of the earth surface restricts the maximum distance between two towers).
  - Mobile phones (900 MHz, 1.8 GHz)
  - Satellite

WIRELESS: MICROWAVE TRANSMISSION

- **Spectrum Allocation**
  - Worldwide, ITU-R coordinates the allocation of spectrum
    - One device can be used in multiple countries
  - In U.S., FCC (Federal Communication Commission) is in charge of the spectrum allocation.
- **Spectrum Allocation Algorithms**
  - Beauty Contest: Each carrier explains why its proposal serves public best.
  - Lottery: Lottery is held among interested companies
  - Auctions: Certain bandwidth is given to the highest bidder.
- **ISM (Industrial, Scientific, Medical) bands**
  - Allocated by government for unlicensed use (device power must be under 1watt to avoid interference)
  - Bluetooth, IEEE 801.11, cordless phone, etc.

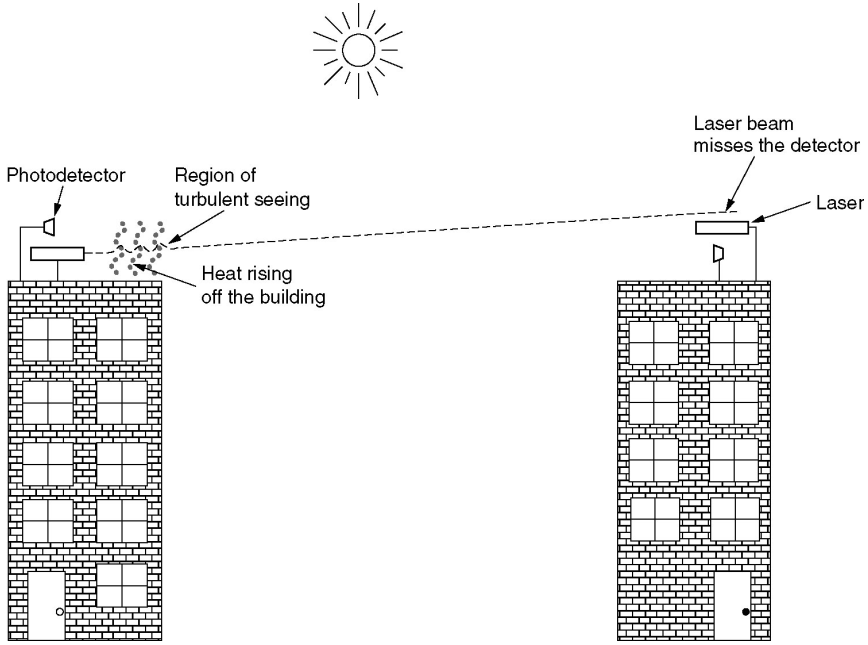


WIRELESS: INFRARED AND MILLIMETER WAVES

- **Characteristics**
  - Short range
  - Directional
  - Do not pass walls
    - As the frequency increases (wavelength decreases), waves behave more like light, less like radio.
    - No interference to adjacent room or building.
  - Doesn't require FCC license
  - Cheap, easy to build
- **Applications**
  - Remote control
  - Connecting PDA and computers
  - Not widely used in communication system.

WIRELESS: LIGHTWAVE TRANSMISSION

- **Characteristics**
  - Line of Sight
  - Unidirectional
  - Absorbed by rain or fog
  - Air turbulence can interfere with communication
- **Applications**
  - Connecting LANs in two buildings via laser



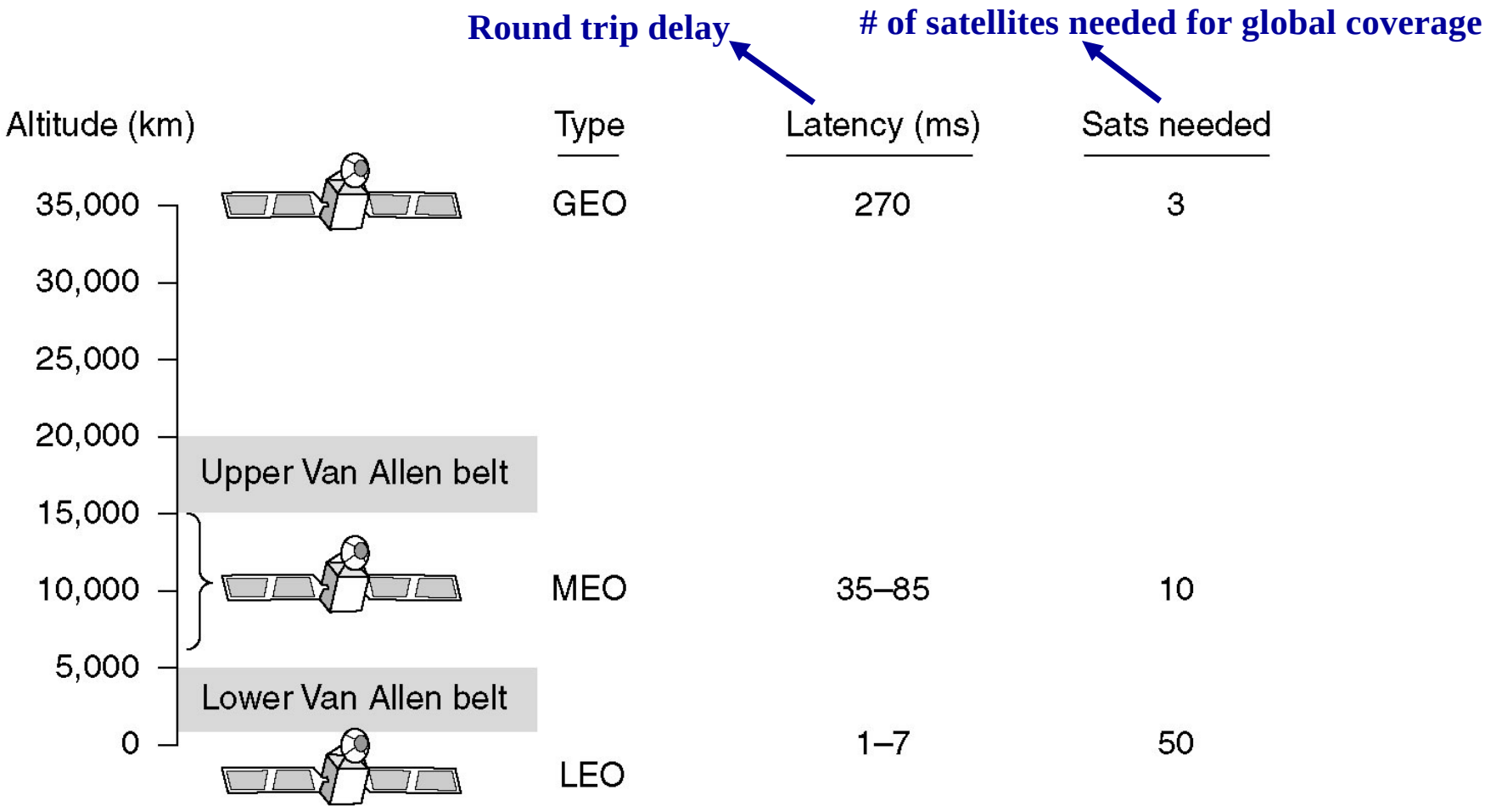
OUTLINE

- Functions of Physical Layer
- Theoretical Basis for Physical Layer Operations
- Guided Transmission Media
- Terrestrial Wireless Transmission
- **Communication Satellite**

COMMUNICATION SATELLITE

- **Classifications based on satellite orbit**
  - Geostationary Earth Orbit (GEO)
    - Orbit Altitude: 35,800 km
    - Period: 24 Hours (Appear to be motionless from earth)
  - Medium Earth Orbit (MEO)
    - Orbit Altitude: 5,000 ~ 15,000 km
    - Period: approx. 6 hours
  - Low Earth Orbit (LEO)
    - Orbit Altitude: 300 ~ 1,500 km
    - Period: approx. 90 minutes

COMMUNICATION SATELLITE



Van Allen belt: layers of highly charged particles, not suitable for satellite.

COMMUNICATION SATELLITE

- **Spectrum: 1.5 GHz – 20 GHz**

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

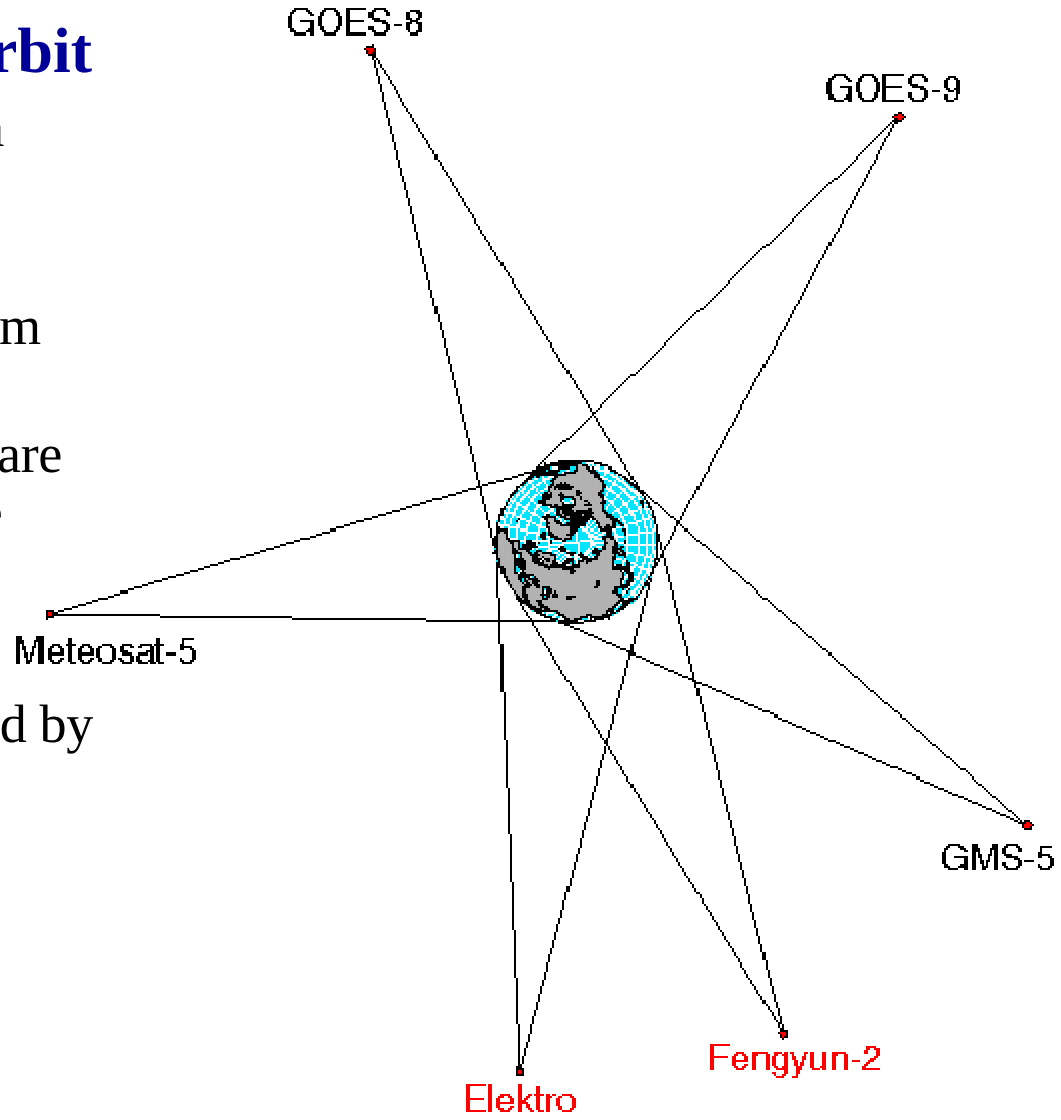
- C-band dish size: 6 – 10 feet.
- Ku-band dish size: 2 – 5 feet (directTV, Dish Network)

COMMUNICATION SATELLITES

- **Characteristics**
  - Relatively long round trip delay
    - GEO: 270 ms
  - Broadcast link
    - Security
    - More suitable for broadcast service: e.g. TV.
  - Able to cover every corner of the entire earth
    - Sea, place with hostile environment, etc.
    - Optic fiber, though with huge bandwidth, is only available to few people.
  - Mobile communication

SATELLITE: GEO

- **Geostationary Earth Orbit**
  - Orbit Latitude: 35,800 km
  - Period: 24 Hours
    - Appear to remain motionless viewed from earth.
  - 3 Geostationary satellites are enough to cover the entire earth.
- **Footprint**
  - The area that is illuminated by a satellite is called the footprint of that satellite.
- **Examples:**
  - DirectTV, dish Network



SATELLITE: MEO

- **Medium Earth Orbit**
  - Orbit Altitude: 5,000 ~ 15,000 km
  - Period: approx. 6 hours
  - 10 satellites are needed to cover the entire earth.

- **Applications**
  - GPS (Global Positioning System)

SATELLITE: LEO

- **Low Earth Orbit**
  - Orbit Altitude: 300 ~ 1,500 km
  - Period: approx. 90 minutes
  - 50 satellites are required to cover the earth.
  - Much lower round trip delay: 1 – 7 ms
    - More suitable for real time communications, such as telephone, Internet.
- **Examples:**
  - Iridium by Motorola (Telephone)
    - 66 Satellites cover the entire earth
    - Launched in 1998, bankrupt in 1999, Sold in 2001
    - Still in operation today
  - Teledisc (Internet, Interactive multimedia)
    - 288 Satellites, later changed to 30 satellites
    - Plan has been suspended.

SUMMARY

- **Physical layer is responsible for Tx information over physical medium.**
- **Theoretical basis**
  - Analog v.s. digital, frequency, bandwidth, capacity, simplex v.s. duplex.
- **Guided transmission media**
  - Removable media, twisted pair, coaxial cable, fiber optics
- **Unguided transmission**
  - Terrestrial wireless transmission
    - Radio transmission, microwave, infrared, unguided light wave
  - Communication Satellite
    - GEO, MEO, LEO

ACKNOWLEDGEMENTS

- **The instructor would like to thank Dr. Ali Kujoory for providing enormous help during the preparation of the course materials and lab experiments.**

- **Part of the slides used in class are based on the instruction materials provided by Dr. Ali Kujoory and Prentice Hall publisher.**